# Emmanuel Atakos *Electrical & Electronics Engineer*


emmanuelatakos@gmail.com

0795628615

Nairobi, Kenya

Emmanuel Atakos


## Profile

Electrical and Electronics Engineer with hands-on experience in electrical installation, maintenance, troubleshooting, and energy monitoring across industrial, commercial, and renewable energy systems. Skilled in High Voltage & low-voltage systems, control panels, wiring, fault diagnosis, preventive maintenance, testing, commissioning support, and technical reporting. Experienced in solar PV design, installation, monitoring, maintenance, IoT gateways and field operations. Currently pursuing Project Management Professional (PMP) certification to strengthen expertise in project planning and management.

## Core Competencies

Electrical Installation (HV & LV Systems)

PV systems Design, Installation & Maintenance

Preventive & Reactive Maintenance (PRM)

Energy Monitoring & Power Systems

Project Management

AutoCAD & Electrical schematics

Technical Reporting & Documentation

Project Coordination

## Professional Experience

2026/06 – Present

**Solar PV Engineer**

*Voltifix New Energy SEZ Ltd Nairobi, Kenya*

- Handle solar PV system design, installation, testing, and commissioning, including panel layout, inverter setup, cabling, protection devices, and system checks.
- Install, commission, configure, and troubleshoot AC and DC electric vehicle (EV) charging stations, ensuring compliance with electrical standards and manufacturer specifications.
- Conduct system inspections and performance assessments while ensuring compliance with technical specifications and safety standards.
- Prepare technical documentation, installation records, and commissioning reports for completed projects.
- Coordinate with engineers, suppliers, and installation teams to ensure timely project delivery.

2024/10 – 2026/05

**Electrical & Automation Engineer**

*Digital Qatalyst Nairobi / Dubai*

- Supported deployment, commissioning, and monitoring of energy systems using smart meters and IoT gateways for real-time electrical consumption tracking.
- Troubleshot electrical and energy system faults to improve reliability, equipment performance, and operational uptime.
- Analysed power and energy consumption data to support fault detection, load behaviour assessment, preventive maintenance, and energy optimisation.
- Prepared technical reports, dashboards, and system performance documentation to support operational and engineering decisions.

2023/01 – 2023/12

**Intern Electrical Engineer**

*Brent Networks Nairobi, Kenya*

- Supported electrical installation, maintenance, and fault-finding activities across operational systems.
- Assisted with preventive maintenance inspections to identify defects, reduce downtime, and support reliable operations.

- Provided solar and renewable energy system support, including monitoring, basic troubleshooting, and performance improvement.
- Conducted electrical testing, documented maintenance observations, and followed safety procedures during field activities and inspections.

2021/01 – 2021/03 **Attachee - Electrical Design**
*Magnum Electrical Technology Nairobi, Kenya*
- Assisted with electrical wiring, lighting, and control system installations in commercial environments, using wiring diagrams to support accurate system layouts.
- Supported circuit efficiency improvements, energy-saving adjustments, and compliance with electrical safety requirements and site procedures.

2020/01 – 2020/03 **Attachee - Electrical Installations**
*Magnum Electrical Technology Nairobi, Kenya*
- Provided hands-on support during installation of electrical wiring, lighting systems, and control panels.
- Identified circuit adjustments that improved system efficiency

2019/09 – 2019/10 **Attachee - Engineering Operations**
*Rivatex East Africa Limited Eldoret, Kenya*
- Supported electrical maintenance technicians in industrial plant operations, assisting with maintenance activities, equipment checks, and safety compliance.
- Prepared electrical schematic and panel layout support drawings using AutoCAD.

## Projects

**Checkout My Portfolio**
*www.emmanuelatakos.vercel.app*

**Solar PV Hybrid Systems**
*Technologies*: *455W / 710W Solar PV modules, Growatt / Jua Power hybrid inverter, lithium battery storage, DC/AC protection, combiner box, smart meter, monitoring system, and AutoCAD.*
- Designed and supported installation of solar PV systems, including system sizing, equipment selection, wiring layout, and protection coordination.
- Supported commissioning, performance monitoring, and maintenance checks to improve system reliability, energy availability, and backup power performance.

**Industry 4.0 Energy Monitoring System**
*Technologies*: *Antigravity, MQTT, Python, Docker, Raspberry Pi, smart energy meters, IoT gateways*
- Designed and implemented a system integrating smart meters, IoT gateways, and dashboards for real-time electrical monitoring
- Simulated and analysed electrical load behaviour for optimisation and fault detection
- Worked with electrical data (power, energy consumption) for system diagnostics

**ESP32 Charge-Point Status Reporter**
*Technologies*: *ESP-IDF, embedded C, FreeRTOS, MQTT, NVS, hardware design, EasyEDA, Wokwi*
- Designed an ESP32-based IoT charger status system for an electric charge point.
- The project simulates a charging station device that: connects to Wi-Fi, sends its live status to the cloud using MQTT, receives remote commands, changes its operating state and shows that state on an RGB LED

**Design and Implementation of a Fiber Optic Network**
*Technologies*: *Optical fiber, network switches, routers, AutoCAD, testing tools*
- Designed network topology, selected equipment, and optimized layouts to improve uptime and increase data speeds by 30%.
- Delivered scalable network infrastructure to support future capacity expansion and technical operations.

## Education

| | |
|---|---|
| 2017 – 2022 | **Bachelor of Science: Electrical & Electronics Engineering**<br>*Jomo Kenyatta University of Agriculture & Technology - Juja, Kenya* |
| 2025 – 2026 | **Project Management Professional (PMP)**<br>*Project Management Institute* |
| 2025 – 2026 | **Data Science**<br>*ALX Africa* |
| 2013 – 2016 | **Kenya Certificate of Secondary Education**<br>*Kakamega School - Kakamega, Kenya* |

## Professional Affiliations

- Graduate Engineer - Institution of Engineers of Kenya (IEK)
- Graduate Engineer (B25236)- Engineers Board of Kenya (EBK)

## References

Available on Request